

LED BAR DISPLAY

Showing Audio Signal Level

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This is a simple audio-signal-based LED indicator built around the LM3914 display driver IC. Visual indication of audio signal levels is an essential feature in audio amplifiers, mixers, music systems, and public-address equipment. The system converts the input audio into a proportional LED bar display, where the number of

illuminated LEDs directly corresponds to the signal amplitude.

The circuit is straightforward to build and well-suited for hobbyists and students. It offers reliable performance with a minimal component count and integrates easily into audio amplifiers, music systems, and PA equipment. The design can be expanded for stereo operation

or higher LED counts by cascading additional LM3914 ICs. Fig. 1 shows the prototype assembled on a breadboard.

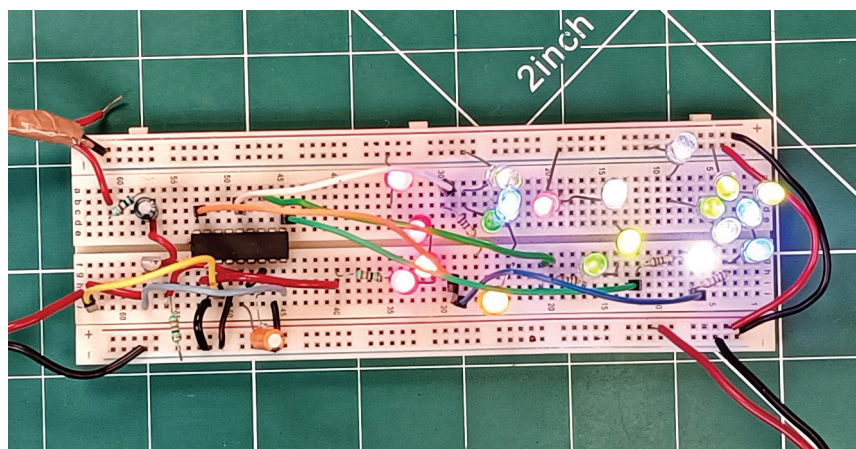


Fig. 1: Author's prototype

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - LM3914N, LED display driver IC
- T1 - BC558, PNP transistor
- BR1 - Bridge rectifier (1A, 400V)
- LED1-LED40 - LEDs, 5mm (any colour)
- LED41 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R4 - 1k Ω
- R5-R14 - 100 Ω

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 2.2 μ F, 25V electrolytic
- C3 - 1 μ F, 25V electrolytic

Miscellaneous:

- X1 - 230V AC to 12V AC step-down Transformer (500mA)
- J1 - 3.5mm audio jack
- PCB - General-purpose or custom built
- Heat sink - For 7805 voltage regulator
- Enclosure, connecting wires, and solder

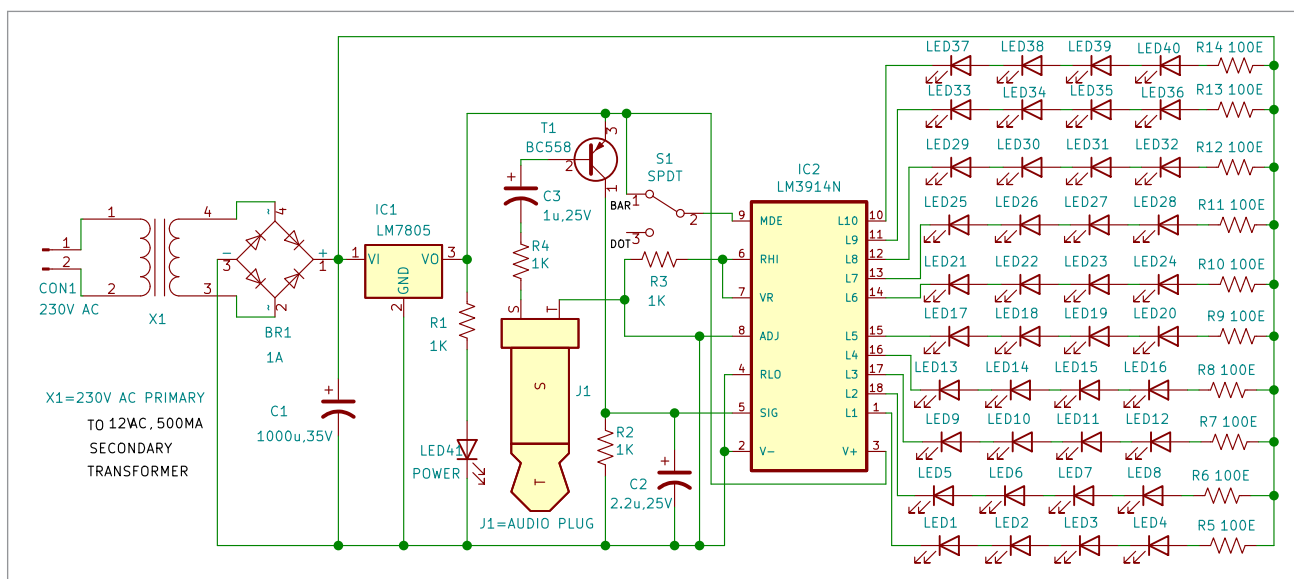


Fig. 2: Circuit diagram